

CUSHING SYNDROME

ACTH-DEPENDENT 90 PERCENT

ACTH-INDEPENDENT 10 PERCENT

CLINICAL PEARL •
LOW ACTH MEANS ADRENAL, HIGH ACTH MEANS PITUITARY OR ECTOPIC

ACTH MOLECULES
ACTH HIGH
75 PERCENT
CUSHING DISEASE
F:M 4:1
PITUITARY PRINCESS
BILATERAL HYPERPLASIA

SMOKING CANNIBISM
ACTH HIGH
15 PERCENT
ECTOPIC SCHEMER
HYPERPIGMENTATION
SMOC LUNG TUMOR
SCLC
THYMUS
MEDULLARY THYROID C-CELL
PNEUROMYOCYTOMA
ADRENALINE DROP
K
HYPOKALEMIC ALKALOSIS
11BETA-HSD2

CHRONIC GLUCOCORTICOID EXCESS
BUFFALO HUMP
CUSHING SYNDROME
CORTISOL
RAPID ONSET
SEVERE HTN
FIRST PROVE CORTISOL, THEN ACTH-SORT IT

ACTH SUPPRESSED
5 TO 10 PERCENT
ADRENAL ADENOMA
F:M 4:1
ADENOMA LADY
STEP 0 EXCLUDE

ACTH SUPPRESSED
1 PERCENT
CARCINOMA KNIGHT
ADRENOCORTICAL CARCINOMA
VIRILIZATION
MALE-BEARD
DHEAS UP
MACRONODULAR
RARE BILATERAL TWINS
EXOGENOUS

1. Glucocorticoid excess king

WHAT YOU SEE

Crowned cushingoid king with buffalo hump on throne, banner 'CHRONIC GLUCOCORTICOID EXCESS'

WHAT IT ENCODES

Cushing SYNDROME = the clinical state of chronic glucocorticoid excess from ANY cause — this throned king is the unifying endpoint every branch in the court feeds into. The buffalo hump, moon face, and central obesity are the shared phenotype. The diagnostic task is two-step: (1) prove cortisol excess biochemically, then (2) find the source. Cushing DISEASE (the pituitary subtype) is just one route to the throne, not the throne itself.

BOARD TRAP

Cushing SYNDROME (the hormonal state, any cause) ≠ Cushing DISEASE (specifically a pituitary ACTH-secreting adenoma). Mixing these two terms is the classic board trap.

2. Exogenous steroids (#1)

WHAT YOU SEE

'EXOGENOUS' banner at bottom right, 'MOST COMMON OVERALL'

WHAT IT ENCODES

Iatrogenic/EXOGENOUS glucocorticoids (prednisone, dexamethasone, inhaled/topical/injected steroids) are by far the #1 cause of Cushing syndrome OVERALL. Because the exogenous steroid suppresses CRH→ACTH, these patients have LOW ACTH and LOW endogenous cortisol, with bilaterally small/atrophic adrenals. They also lack hyperpigmentation (low ACTH) and often lack the androgenic features (low adrenal androgens). Always take a meticulous drug history first.

BOARD TRAP

The most common OVERALL cause is exogenous steroids, NOT a pituitary adenoma. Pituitary adenoma is only the most common cause of ENDOGENOUS Cushing. Read which one the stem is asking.

3. ACTH-dependent 90%

WHAT YOU SEE

Left banner 'ACTH-DEPENDENT 90 PERCENT', 'ACTH HIGH'

WHAT IT ENCODES

Of ENDOGENOUS Cushing, ~80-90% is ACTH-dependent — the adrenal is being driven by high or inappropriately normal ACTH. This causes bilateral adrenal hyperplasia. The two sources are pituitary (Cushing disease, the large majority) and ectopic ACTH. High ACTH also drives MC1R → hyperpigmentation, a clue absent in adrenal/exogenous causes.

BOARD TRAP

'ACTH-dependent' refers to ENDOGENOUS cases only; don't apply the 90% figure to all Cushing. And high/normal ACTH still requires further testing (high-dose dex, CRH, IPSS) to separate pituitary from ectopic.

4. Pituitary princess (Cushing disease)

WHAT YOU SEE

Princess in pink, '75 PERCENT', 'CUSHING DISEASE', 'BILATERAL HYPERPLASIA'

WHAT IT ENCODES

The pituitary corticotroph (ACTH-secreting) adenoma = Cushing DISEASE — the single most common cause of ENDOGENOUS Cushing (~70-80%), usually a small microadenoma, classically in a young woman (note F:M ~4:1). It causes bilateral adrenal hyperplasia. It retains partial feedback: SUPPRESSES with high-dose dexamethasone and RESPONDS to CRH. First-line treatment is transsphenoidal resection.

BOARD TRAP

Cushing DISEASE = pituitary source ONLY; it is not synonymous with Cushing syndrome. Also, the pituitary lesion is often a tiny microadenoma that MRI can MISS — a normal pituitary MRI does not exclude Cushing disease (use IPSS).

5. Ectopic schemer

WHAT YOU SEE

Dark-suited man with SCLC/carcinoid tumor, '15 PERCENT', 'SMOKING'

WHAT IT ENCODES

Ectopic ACTH (~10-15% of endogenous) — a non-pituitary tumor secreting ACTH: small-cell lung carcinoma (most common, smoker), bronchial/thymic carcinoid, medullary thyroid carcinoma, pheochromocytoma, pancreatic NET. Presentation is often RAPID with very high ACTH and cortisol, severe hypertension, hypokalemic metabolic alkalosis, weight loss, and prominent hyperpigmentation — frequently WITHOUT the classic slow Cushingoid habitus (no time to develop).

BOARD TRAP

Ectopic ACTH does NOT suppress with high-dose dexamethasone and does NOT respond to CRH (the opposite of pituitary disease) — but bronchial carcinoid is a notorious exception that can mimic pituitary behavior, which is exactly why IPSS exists to settle it.

6. Hypokalemic alkalosis

WHAT YOU SEE

11-beta-HSD2 enzyme arrow near ectopic, 'HYPOKALEMIC ALKALOSIS'

WHAT IT ENCODES

Mechanism of the electrolyte picture in severe (esp. ectopic) Cushing: normally renal 11β-HSD2 inactivates cortisol to cortisone, protecting the mineralocorticoid receptor. Massive cortisol SATURATES 11β-HSD2, so cortisol floods the mineralocorticoid receptor → Na/water retention, K and H wasting → HYPOKALEMIC METABOLIC ALKALOSIS and hypertension. This is most marked in ectopic ACTH because cortisol levels are the highest.

BOARD TRAP

This is the same mechanism by which licorice (glycyrrhetic acid inhibits 11β-HSD2) mimics mineralocorticoid excess — 'apparent mineralocorticoid excess.' Hypokalemic alkalosis with high cortisol points to a severe/ectopic source rather than a mild pituitary one.

7. ACTH-independent 10%

WHAT YOU SEE

Right banner 'ACTH-INDEPENDENT 10 PERCENT', 'ACTH SUPPRESSED'

WHAT IT ENCODES

~10-20% of endogenous Cushing is ACTH-INDEPENDENT — the adrenal makes cortisol autonomously, so pituitary ACTH is SUPPRESSED (low). Causes: adrenal adenoma (most common), adrenal carcinoma, and rare bilateral conditions (macronodular hyperplasia, PPAD/Carney). Because ACTH is low, the contralateral and non-tumor adrenal tissue atrophies — a key reason to taper/replace steroids after unilateral adrenalectomy.

BOARD TRAP

Low ACTH = adrenal source — go straight to adrenal CT, do NOT do high-dose dex or CRH testing (those are for sorting ACTH-DEPENDENT cases). Ordering pituitary tests in a low-ACTH patient is a classic wasted-step trap.

8. Adenoma lady

WHAT YOU SEE

Girl labeled 'ADRENAL ADENOMA', '5 to 10 PERCENT', 'STEP 0 EXCLUDE'

WHAT IT ENCODES

Benign cortisol-secreting ADRENAL ADENOMA — the most common ACTH-INDEPENDENT cause. Typically a small (<4 cm), homogeneous, lipid-rich (<10 HU) unilateral mass secreting cortisol ALONE (pure phenotype, no virilization). Treatment is unilateral adrenalectomy; expect transient adrenal insufficiency afterward because the suppressed contralateral gland is atrophic and needs glucocorticoid cover with a taper.

BOARD TRAP

A solitary adenoma secretes ONE hormone (cortisol only); when you see MIXED cortisol + androgen excess or a large heterogeneous mass, switch your answer to adrenocortical CARCINOMA, not adenoma.

9. Carcinoma knight (ACC)

WHAT YOU SEE

Armored knight '1 PERCENT', 'GREATER THAN 4 CM', '>20 HU', 'HETEROGENEOUS', virilization/deep voice

WHAT IT ENCODES

Adrenocortical CARCINOMA — rare (~1% of Cushing) but aggressive. Imaging red flags: size >4 cm, unenhanced attenuation >20 HU (lipid-poor), heterogeneity/necrosis/calcification, slow contrast washout. Biochemistry red flag: MIXED steroid excess — cortisol PLUS androgens (virilization, deep voice, hirsutism) ± mineralocorticoid. Rapid-onset Cushing + virilization + large mass = ACC; treat with complete resection ± mitotane.

BOARD TRAP

Don't be reassured that a mass is 'functioning' — function does not equal benign. Size >4 cm, high HU, heterogeneity, and mixed-hormone output distinguish ACC from a benign adenoma.

10. Rare bilateral twins

WHAT YOU SEE

Twin babies, 'PRIMARY PIGMENTED NODULAR', 'CARNEY COMPLEX', 'MACRONODULAR'

WHAT IT ENCODES

Rare ACTH-independent BILATERAL disease: (1) Primary Pigmented Nodular Adrenocortical Disease (PPAD) — small pigmented nodules, part of CARNEY complex (PRKAR1A mutation; also myxomas, lentigines/spotty skin pigmentation, schwannomas), often young patients with a paradoxical RISE in cortisol on dexamethasone. (2) Bilateral Macronodular Adrenal Hyperplasia (BMAH/ARMC5) — large nodules, sometimes driven by aberrant GPCRs (food/GIP-dependent, vasopressin, etc.).

BOARD TRAP

Both are BILATERAL with LOW ACTH — so a low ACTH does not always mean a single unilateral adenoma; bilateral adrenal enlargement with suppressed ACTH should make you think macronodular hyperplasia or PPAD/Carney.

11. Prove cortisol first scale

WHAT YOU SEE

Balance scale 'FIRST PROVE CORTISOL THEN ACTH SORT IT'

WHAT IT ENCODES

The workup ALGORITHM in one image: STEP 1 — confirm hypercortisolism with ≥2 first-line screens (1 mg overnight dexamethasone suppression, late-night salivary cortisol, or 24-h urine free cortisol). STEP 2 — only THEN measure plasma ACTH to localize. Step 0 (the 'EXCLUDE' tag) is always ruling out exogenous steroids first. Imaging comes LAST, after biochemistry tells you where to look.

BOARD TRAP

Never image the adrenals or pituitary BEFORE biochemical confirmation — incidentalomas are common (~5% of CTs) and will mislead you. Also never check ACTH before you have proven cortisol excess.

12. Clinical pearl ACTH rule

WHAT YOU SEE

'LOW ACTH MEANS ADRENAL, HIGH ACTH MEANS PITUITARY OR ECTOPIC'

WHAT IT ENCODES

The master localizing rule: LOW (suppressed) ACTH → ACTH-INDEPENDENT → ADRENAL source → adrenal CT. HIGH/normal ACTH → ACTH-DEPENDENT → PITUITARY or ECTOPIC → then use high-dose dexamethasone + CRH stimulation (pituitary suppresses & responds; ectopic does neither) and IPSS to break ties. Mnemonic: 'Low ACTH = look Low (adrenal); High ACTH = look High (pituitary) or hidden (ectopic).'

BOARD TRAP

Inappropriately NORMAL ACTH in a patient with proven cortisol excess counts as ACTH-DEPENDENT, not normal — autonomous adrenal disease should have FULLY suppressed it. Treat a non-suppressed ACTH as 'high.'